

ECO 310: Precept 1

1 Transitivity

Question: A 3-committee uses majority rule to determine their preferences. That is, for any choices x and y , the committee prefers x to y iff at least 2 of the 3 committee members prefer x to y . Show that the committee preferences may not be transitive.

Solution: For example, choose 3 alternatives x, y, z . And consider the following individual preferences, which yield social preference $x \succ y$ (thanks to people 1 and 3) and $y \succ z$ (thanks to people 1 and 2). Transitivity would require $x \succ z$. But this is not the case, since people 2 and 3 both prefer z to x .

$$\begin{array}{ll} 1 & : \quad x \succ y \succ z \\ 2 & : \quad y \succ z \succ x \\ 3 & : \quad z \succ x \succ y \end{array}$$

2 Lexicographic Preferences

Consider lexicographic preferences on $\mathbb{R}^2$; i.e., $(x_1, x_2) \succsim (y_1, y_2)$ if $x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$.

1. Draw an indifference “curve” through the point $(3, 2)$.

Answer: it is just this single point, since this DM is never indifferent between distinct bundles: If one bundle has more of good 1, it’s automatically strictly preferred. And if the two bundles have the same amounts of good 1, then the one with more good 2 is strictly preferred. Indifference only happens when quantities of both goods are equal, i.e. between identical bundles. (Recall that $x \sim y$ requires weak preference in both directions, $x \succsim y$ and $y \succsim x$).

2. Find the best affordable bundle if each good costs \$2/unit and the consumer has \$5.

Answer: with lexicographic preferences, among any two bundles, the DM always prefers the one with more of good 1. So it is never optimal to spend money on good 2, i.e. to buy a bundle (x_1, x_2) with $x_2 > 0$, as if he reallocated the money to good 1, he’d move to a bundle with more good 1, i.e. that is strictly preferred.

So an optimizing consumer will spend everything on good 1; with \$5, this lets him buy the bundle $(2.5, 0)$.

3. Prove that lexicographic preferences are strictly monotonic.

Answer: we want to show that getting more of either good, without losing anything, makes the consumer strictly better off. To check this, consider an initial bundle (x_1, x_2) . If we increase just the amount of good 1 by say ε , we get a new bundle $(x_1 + \varepsilon, x_2)$, which is strictly preferred to (x_1, x_2) since it has more good 1. If we increase just the amount of good 2 by say ε , we get a new bundle $(x_1, x_2 + \varepsilon)$, which is again strictly preferred to (x_1, x_2) since these two bundles have the same amount of good 1, but the new one has more good 2.

3 Choices

I suggested in class constructing a preference from choices by: $x \succsim y$ if $x \in C(\{x, y\})$ (i.e. declaring x preferred to y if you'd be willing to pick x when these are the only options). And claimed that if your actual choices satisfy Sen's α and β axioms, then for any menu A , if x is $\succsim$ -optimal, then $x \in C(A)$. Prove this.

Answer: suppose, toward a contradiction, that there is a set A and an option $x \in A$ which is $\succsim$ -optimal, but which you don't choose. Then:

- since choice sets are not allowed to be empty, there must be some other $z \in A$ with $z \in C(A)$. But by IIA, if z is chosen from A , then it must also be chosen from any submenu of A where it's an option, in particular from $\{x, z\}$. So $z \in C(\{x, z\})$; by construction, this implies $z \succsim x$.
- By assumption, x is $\succsim$ -optimal in A , thus x is weakly preferred to every option in A , including the z from the above bullet point: so in particular $x \succsim z$.
- From the first two bullet points, $x \sim z$, and so by construction of $\succsim$, this means x and z are both chosen from $\{x, z\}$
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- ut then by Sen's β axiom, if $\{x, z\}$ are both chosen from $\{x, z\}$, and (from first bullet point) z is chosen from A —a larger menu containing $\{x, z\}$ —then x must also be chosen from A . A contradiction.

4 Marginal Rate of Substitution

1. Consider the utility function $u(C, T) = (4C + T)^2$. Explain: this person is always willing to trade one coffee for four teas.

Answer: from Lecture 2,

$$MRS_{\text{coffee for tea}} = - \left. \frac{\Delta \text{coffee}}{\Delta \text{tea}} \right|_{\text{indifference}} = \frac{MU_{\text{tea}}}{MU_{\text{coffee}}}$$

In this example, you might notice that the initial utility function represents the same preferences as simply $v(C, T) = 4C + T$ does, since utility functions u and v are monotone transformations of each other. And here, you can simply read off that the marginal utility of coffee (partial derivative of v in C) is 4, while the marginal utility of tea (partial derivative of v in T) is 1. So $\frac{MU_{\text{tea}}}{MU_{\text{coffee}}} = \frac{1}{4}$, as desired: he'll trade coffee for tea at a 1-for-4 rate. Had you not transformed the utility function, you would still have gotten:

$$\begin{aligned} MU_{\text{coffee}} &= \frac{\partial u}{\partial C} = 2(4C + T) \cdot 4 \\ MU_{\text{tea}} &= \frac{\partial u}{\partial T} = 2(4C + T) \cdot 1 \\ \Rightarrow \frac{MU_{\text{tea}}}{MU_{\text{coffee}}} &= \frac{2(4C + T) \cdot 1}{2(4C + T) \cdot 4} = \frac{1}{4} \end{aligned}$$

2. If time left over: explain why if $p_C < 4p_T$, then this consumer should spend all his money on coffee.

Answer: here, coffee always has 4 times the value of tea ($\frac{MU_C}{MU_T} = 4$). So if it is less than 4 times as expensive, then it always has more “bang-per-buck” i.e. utility-boost per dollar. ($\frac{MU_C}{p_C} > \frac{MU_T}{p_T}$). So a utility-maximizing consumer should spend all their money on coffee.